



COURSE SYLLABUS

CS201 – Computer Systems Programming

1. GENERAL INFORMATION

Course name:	Computer Systems Programming
Course name (in Vietnamese):	Kiến trúc máy tính và Hợp ngữ
Course ID:	CS201
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	CS161, CS162
Prior-course:	
Instructors:	Dinh Dien

2. COURSE DESCRIPTION

The course is an introduction to computer systems from a software perspective. Topics include: Basic machine organization; System programming using C and assembly language; Introduction to system programming tools (gcc, makefile, gdb); Data representation (bits & bytes, characters, integers, floating point numbers), Implementation of control flow, procedure class, and complex data types at machine level; Linking and loading; Exceptions and interrupts; Process control and signals; System calls, File I/O; Timing and improving program performance; Introduction to memory hierarchy, dynamic memory allocation techniques.

3. COURSE GOALS

At the end of the course, students are able to use their knowledge about data presentation, assembly language, memory and others to write better program. Students also know how things work, the security holes, debugging without source code and how to fix them when they break. At the end of the course, students are able to:

ID	Description	Program LOs
G1	Explain common bit-level representations of numeric values (unsigned, two's complement, floating point) and the consequent mathematical properties of arithmetic and bit level operations on them	1, 2
G2	Recognize the relation between programs expressed in C and in assembly code, including the implementation of expressions, control, procedures, and data structures	3, 4
G3	Demonstrate ability to understand basic intention of a program through its binary representation and apply these skills to debugging programs	6, 7
G4	Explain the concepts of computer memory and how it works, analyze the consequences of imperfect system usage, such as poor memory and CPU performance and apply to write better and faster programs	8, 9, 10

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Explain common bit-level representations of numeric values (unsigned, two's complement, floating point) and the arithmetic and bit level operations (and, or, xor, not, shift,...)	I
G1.2	Implement programs using bit-operations	T

G2.1	Understand machine architecture and language including the basic concept, control code, procedures, array, heterogeneous data structure	I
G2.2	Read the machine code generated by a C compiler and write a simple Linux shell by assembly language	T
G3.1	Understand the basic intention of a program through its binary representation and the common code security vulnerabilities, such as buffer overflow	T
G4.1	Explain the concepts of computer memory and how these storage devices are arranged in a hierarchy	I
G4.2	Simulate cache memory and improve the performance of application programs by improving their temporal and spatial locality	T

5. TEACHING PLAN

5.1 Theory:

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Introduction Bits and Bytes	G1.1, G1.2	Lecturing (Chapter 1.1-1.10, 2.1)
2	Integers and Floating points	G1.1, G1.2	Lecturing (Chapter 2.2-2.4)
3	Machine Programming: Basics and Control	G2.1	Lecturing (Chapter 3.1-3.6)
4	Machine Programming: Procedures	G2.1	Lecturing (Chapter 3.7)
5	Review Midterm Exam		Q&A, Discussion, Exercises

6	Machine Programming: Array and data	G2.1, G2.2	Lecturing (Chapter 3.8-3.9)
7	Machine Programming: Advanced topics	G2.2, G3.1	Lecturing (Chapter 3.9-3.12)
8	Memory Hierarchy	G4.1	Lecturing (Chapter 6.1-6.3)
9	Cache Memories	G4.1, G4.2	Lecturing (Chapter 6.4-6.6)
10	Cache Memories (con.) Virtual Memory (optional)	G4.1, G4.2	Lecturing (Chapter 6.4-6.6, 9.1-9.6)
11	Review		Q&A, Discussion, Exercises

5.1 Laboratory:

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded.

Week	Topic	Courses Outcomes	Programming assignments
1, 2,3	Information Representation: Bits and Bytes, Integers and Floating points	G1.1, G1.2	Datalab
4, 5, 6	Machine Programming: Basics and Control	G2.1, G2.2	Binary bomb lab
7, 8, 9	Advanced Machine Programming: Procedures, Data Structures	G2.2, G3.1	Buffer overflow lab
10	Review	G1.1, G1.2, G2.1 G2.2 G4.1, G4.2	

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Homework	Class material is provided by prof. DDien in class and on the Moodle	G1.1, G2.1, G3.1, G4.1	20%
A2	Programming assignments (Laboratory)			20%
A21	Data lab (W1-2)	Implement simple logical and arithmetic C functions using bit-level operations	G1.1, G1.2	6%
A22	Binary bomb lab (W3-5)	Each student receives an unique machine code file (bomb), and try to defuse the bomb by disassembling and reverse engineering the C program to determine what the answer should be	G2.1, G2.2	7%
A23	Buffer overflow lab (W6-8)	Modify the run-time behavior of a binary executable by exploiting a buffer overflow vulnerability	G2.2, G3.1	7%
A3	Exams			60%
A31	Midterm exam	Written (closed book) The midterm exam will cover the basic concepts involved in the data representation (bits bytes, integers, floats, and bit level operations) and machine level representation of	G1.1, G1.2 G2.1, G2.2	20%

		programs (basics, conditions, loops)		
A33	Final exam	Written (closed book) The final exam will cover the advanced concepts involved in advanced machine level representation of programs (procedures, data structures) and the memory hierarchy	G2.1, G2.2 G3.1 G4.1, G4.2	40%

7. RESOURCES

Textbooks

- *Computer Systems: A programmer's Perspective* by Bryant O'Hallaron (main)

Others

- *Digital Design and Computer Architecture* by David Money Harris & Sarah L.Harris (reference)

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.